1.

a	*b*	*c*	*abc*
1	**10**	**4**	**40**
2	**6**	**7**	**84**
3	**2**	**10**	**60**
All values of *abc* are multiples of 4			

One correct set *a*, *b* & *c*	B1
Two correct rows	M1
Fully correct + statement	A1

(3 marks)

2.(a)	$u_2 = \frac{2}{3},\ u_3 = -4,\ u_4 = 3$	M1, A1, A1 **(3)**
(b)	$\sum_{r=1}^{100} u_r = 33 \times \left(3 + \frac{2}{3} + -4\right) + 3$	M1
	$= -8$	A1 **(2)**
		(5 marks)

	$2\tan\theta + 3\sin\theta = 0$	
3 (a)	States or uses $\tan\theta = \dfrac{\sin\theta}{\cos\theta} \rightarrow 2\dfrac{\sin\theta}{\cos\theta} + 3\sin\theta = 0$	M1
	$\sin\theta\left(2 + 3\cos\theta\right) = 0$	
	$2 + 3\cos\theta = 0$	dM1
	$\cos\theta = -\dfrac{2}{3} \Rightarrow \theta =$ awrt $132°$ or awrt $228°$	A1
	$\cos\theta = -\dfrac{2}{3} \Rightarrow \theta =$ awrt $131.8°$ and awrt $228.2°$	A1
	$\sin\theta = 0 \Rightarrow \theta = 180°, 360°$	B1
		(5)
(b)	Sets $2x + 40° =$ their $131.8° \Rightarrow x = ...$	M1
	$x =$ awrt $45.9°$	A1
		(2)
		(7 marks)

	$f(x) = 4x^3 + ax^2 - 29x + b$	
4.(a)	Sets $f\left(-\frac{1}{2}\right) = 0 \rightarrow 4\times\left(-\frac{1}{2}\right)^3 + a\times\left(-\frac{1}{2}\right)^2 - 29\times\left(-\frac{1}{2}\right) + b = 0$	M1
	$\Rightarrow \frac{1}{4}a + b + 14 = 0 \Rightarrow a + 4b = -56$ *	A1*
		(2)
(b)	Sets $f(2) = -25 \rightarrow 4\times 2^3 + a\times 2^2 - 29\times 2 + b = -25$	M1
	$4a + b = 1$	A1
		(2)
(c)	(i) Solves $a + 4b = -56$ with their $4a + b = 1 \Rightarrow a = ..., b = ...$	M1
	$a = 4, b = -15$	A1
	(ii) $4x^3 + 4x^2 - 29x - 15 = (2x+1)\left(2x^2 + x - 15\right)$	M1, A1
	$= (2x+1)(2x-5)(x+3)$ cso	A1
		(5)
		(9 marks)

5. (i)	$3^a = 70 \Rightarrow a\log 3 = \log 70$	M1
	$\Rightarrow a = \dfrac{\log 70}{\log 3} = 3.867$	A1
		(2)
(ii)	$4 = \log_3 81 \quad \text{or} \quad 3\log_3 b = \log_3 b^3 \quad \text{or} \quad \log_3 5b = \log_3 5 + \log_3 b$	B1
	$4 + 3\log_3 b = \log_3 5b \Rightarrow \log_3 81b^3 = \log_3 5b$	M1
	$\Rightarrow 81b^{3} = 5b \Rightarrow b = \ldots$	dM1
	$\Rightarrow b = \sqrt{\dfrac{5}{81}}$	A1
		(4)
		(6 marks)

6. (a)	Identifies $h = 3$	B1
	Area $= \frac{3}{2}\{0+0+2(1.52+2.74+3.12+3.08)\}$	M1
	$= 31.38\ \text{m}^2$	A1
		(3)
(b)	Calculates $"31.38"\times 1.5\times 60$	M1
	$=$ awrt $2800\ \text{m}^3$	A1
		(2)
(c)	Underestimate with valid reason, e.g. area of trapezia are smaller than cross sectional area	B1
		(1)
		(6 marks)

7(a)	States or implies that gradient $XN = -\frac{2}{5}$	B1
	Uses $(4,-3)$ with a gradient of $\pm\frac{2}{5}$ to form equation of XN	M1
	$y+3=-\frac{2}{5}(x-4)$	
	$5y+15=-2x+8 \Rightarrow 2x+5y+7=0$ *	A1*
		(3)
(b)	(i) Solves $2x+5y+7=0$ and $y=\frac{5}{2}x-\frac{55}{2} \Rightarrow x=...,y=...$	M1
	$N=(9,-5)$	A1
	(ii) Attempts $("9"-4)^2+("-5"+3)^2$	M1
	Correct form of equation E.g. $(x-4)^2+(y+3)^2=("9"-4)^2+("-5"+3)^2$	M1
	$(x-4)^2+(y+3)^2=29$	A1
		(5)
		(8 marks)

8. (a)	Uses $S_{10} = 360 \Rightarrow \frac{10}{2}\{2\times a + 9\times 4\} = 360$ $\Rightarrow 2a + 36 = 72 \Rightarrow a = 18$	M1 A1 **(2)**
(b)	Uses $S_n = 2146 \Rightarrow \frac{n}{2}\{2\times "18" + (n-1)\times 4\} = 2146$ $\Rightarrow n\{16 + 2n\} = 2146$ $\Rightarrow n^2 + 8n - 1073 = 0$ *	M1 A1* **(2)**
(c)	(i) States 29 (ii) Attempts $"18" + ("29" - 1)\times 4 = 130$	B1 M1, A1 **(3)** **(7 marks)**

9 (a)	$y = \frac{2}{3}x^{2} - 9\sqrt{x} + 13$ $\frac{dy}{dx} = \frac{4}{3}x - \frac{9}{2\sqrt{x}}$	M1, A1
	Attempts to solve $\frac{dy}{dx} > 0 \Rightarrow x^{\frac{3}{2}} > \frac{27}{8} \Rightarrow x > \frac{9}{4}$	dM1, A1 **(4)**
(b)	$\int \frac{2}{3}x^{2} - 9\sqrt{x} + 13\,dx = \frac{2}{9}x^{3} - 6x^{\frac{3}{2}} + 13x \quad (+c)$	M1
	Area $= \left[\frac{2}{9}x^{3} - 6x^{\frac{3}{2}} + 13x\right]_{0}^{9} = 117$	dM1, A1
	Substitutes $x = 9$ into $\frac{dy}{dx} = \frac{4}{3}x - \frac{9}{2\sqrt{x}} \Rightarrow \frac{dy}{dx} = \left(\frac{21}{2}\right)$	M1
	Finds where tangent meets x-axis $0 - 40 = \frac{21}{2}(x - 9) \Rightarrow x = \frac{109}{21}$	dM1 A1
	Area of R $= \left[\frac{2}{9}x^{3} - 6x^{\frac{3}{2}} + 13x\right]_{0}^{9} - \frac{1}{2} \times 40 \times \left(9 - \frac{109}{21}\right)$	ddM1
	$= 117 - \frac{1600}{21} = \frac{857}{21}$	A1 **(8)** **(12 marks)**

10.(i)(a)	$(3+2x)^6$	
	Any correct form for any term of ${}^6C_1 3^5 (2x)^1$ or ${}^6C_2 3^4 (2x)^2$ or ${}^6C_4 3^2 (2x)^4$	M1
	Any two correct: t(2) = $6\times 3^5\times(2x)^1$, t(3) =$15\times 3^4\times(2x)^2$ t(5) = $15\times 3^2\times(2x)^4$	A1
	All 3 terms correct 2nd term = 2916x, 3rd term= 4860x^2 , 5th term=2160x^4	A1
		(3)
(b)	Method using consecutive terms of a GP e.g $\dfrac{2160x^4}{4860x^2}=\dfrac{4860x^2}{2916x}$	M1
	$\Rightarrow ...x=...$	dM1
	$\Rightarrow x=\dfrac{15}{4}$	A1
		(3)
(ii) (a)	Attempts to use $S_\infty=\dfrac{a}{1-r}\Rightarrow\dfrac{8}{5}=\dfrac{\sin^2\theta}{1-2\cos\theta}$	M1
	$\Rightarrow\dfrac{8}{5}=\dfrac{1-\cos^2\theta}{1-2\cos\theta}$	dM1
	$\Rightarrow 8-16\cos\theta=5-5\cos^2\theta$	
	$\Rightarrow 5\cos^2\theta-16\cos\theta+3=0$	A1*
		(3)
(b)	Solves $5\cos^2\theta-16\cos\theta+3=0\Rightarrow\cos\theta=\dfrac{1}{5}$	B1
	Attempts $ar=\sin^2\theta\times 2\cos\theta=\left(1-\left(\dfrac{1}{5}\right)^2\right)\times 2\times\dfrac{1}{5}$	M1
	$=\dfrac{48}{125}$	A1
		(3)
		(12 marks)